SAMPLE QUESTION PAPER

SESSION 2025-2026

Class 11th

PHYSICS

SET-03

Hints & Solutions

1. (A)

(i) (C) The trailing zeros in a number are significant if they are to the right of the decimal point. In 5.00, both zeros are to the right of the decimal point and are significant.

(ii) (A) u/g. Using the first equation of motion, $v = u + at$. At the maximum height, the final velocity (v) is 0. The acceleration (a) is due to gravity and is negative ($a = -g$).

$0 = u - gt$

$\Rightarrow t = u/g$

(iii) (B) Newton's second law of motion. The work-energy theorem, which states that the work done on a body equals the change in its kinetic energy, is derived directly from Newton's second law of motion $(F = ma)$.

(iv) (D) Gravitational force. A force is conservative if the work done by it in moving a body between two points is independent of the path taken. The work done by gravitational force depends only on the initial and final positions.

(v) (D) Obeys both the first and second laws of thermodynamics. A refrigerator uses external work to transfer heat from a cold reservoir to a hot reservoir, which is a process allowed by both the first law (conservation of energy) and the second law of thermodynamics.

(vi) (B) The net torque on the particle about that point is zero. The principle of conservation of angular momentum states that if the net external torque (τ) acting on a system is zero, the angular momentum (L) of the system remains constant. Mathematically, $\tau = \frac{dL}{dt}$. If $\tau = 0$, then $\frac{dL}{dt} = 0$, which means L is a constant.

(vii) (A) $M^1L^0T^{-2}$. Surface tension (γ) is defined as force per unit length.

Force, $F = M^1L^1T^{-2}$

Length, $L = L^1$

Therefore, the dimensional formula for surface tension is:

$\gamma = \frac{F}{L} = \frac{M^1L^1T^{-2}}{L^1} = M^1L^0T^{-2}$.

1. **(B)**

(i) The principle of homogeneity of dimensions states that an equation is dimensionally correct if the dimensions of all the terms on both sides of the equation are the same.

(ii) Centripetal acceleration is the acceleration of a body moving in a circular path, which is directed towards the center of the circle.

(iii) The law of conservation of energy states that energy can neither be created nor destroyed, it can only be converted from one form to another. The total energy of an isolated system remains constant.

(iv) The area under a velocity-time graph represents the displacement of the object.

(v) The Zeroth Law of Thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third one, then they are in thermal equilibrium with each other.

(vi) The relationship is $\gamma = 3\alpha$, where α is the coefficient of linear expansion and γ is the coefficient of cubical expansion.

(vii) A geostationary satellite is a satellite that orbits the Earth at the same rate as the Earth's rotation, remaining at a fixed position relative to a point on the Earth's surface.

2. (a)

- Streamline Flow: A flow in which the fluid velocity at any point is constant over time. The path of each fluid particle is a streamline, and no two streamlines cross each other.
- Turbulent Flow: A flow characterized by erratic, chaotic, and random motion of fluid particles. The flow velocity at any point changes erratically.

OR

The principle of superposition of waves states that when two or more waves overlap, the resulting displacement at any point is the algebraic sum of the individual displacements of the waves.

3. Parallelogram Law of Vector Addition: If two vectors acting simultaneously at a point are represented by the two adjacent sides of a parallelogram drawn from that point, then the resultant vector is represented by the diagonal of the parallelogram drawn from the same point.
Derivation:
Let A and B be two vectors inclined at an angle θ. The resultant vector R is the diagonal of the parallelogram. Using the law of cosines in the triangle formed by the vectors and the resultant, the magnitude of the resultant is:

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

The direction of the resultant with respect to vector A is given by the angle α :

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

4. Consider a satellite of mass m orbiting a planet of mass M in a circular orbit of radius r . The necessary centripetal force is provided by the gravitational force.

Gravitational Force: $F_g = \frac{GMm}{r^2}$

Centripetal Force: $F_c = \frac{mv^2}{r}$

Equating the two forces:

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$\Rightarrow v^2 = \frac{GM}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

This is the expression for the orbital velocity.

OR

[2]

Work-Energy Theorem: The work done by the net force on an object equals the change in its kinetic energy.

Proof for a variable force:

Work done on a particle by a variable force is $W = \int_{x_1}^{x_2} F dx$.

From Newton's second law, $F = ma = m\frac{dv}{dt}$.

$$W = \int_{x_1}^{x_2} m\frac{dv}{dt}dx = \int_{v_1}^{v_2} m\frac{dv}{dx}\frac{dx}{dt}dv = \int_{v_1}^{v_2} mvdv$$

$$= m\left[\frac{v^2}{2}\right]_{v_1}^{v_2} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta KE.$$

This proves the work-energy theorem.

5. First Law of Thermodynamics: $\Delta U = Q - W$, where ΔU is the change in the internal energy of the system, Q is the heat supplied to the system, and W is the work done by the system.

Explanation: The law states that the heat supplied to a system is used for two purposes: to increase the internal energy of the system and to do work on its surroundings. It is a statement of the law of conservation of energy for a thermodynamic system.

6.

- Moment of Inertia: It is the rotational equivalent of mass. It measures the resistance of a body to a change in its rotational motion.
- Parallel Axis Theorem: $I = I_{cm} + Md^2$, where I is the moment of inertia about an axis parallel to the axis through the center of mass, I_{cm} is the moment of inertia about the center of mass, M is the mass, and d is the distance between the two axes.
- Perpendicular Axis Theorem: For a planar body, the moment of inertia about an axis perpendicular to its plane is the sum of the moments of inertia about two perpendicular axes in the plane, all three axes being concurrent at a single point. $I_z = I_x + I_y$.

7. Change in Internal Energy (ΔU): For a cyclic process, the system returns to its initial state. The internal energy is a state function, meaning its value depends only on the state of the system. Therefore, the change in internal energy for a complete cycle is zero.

$\Delta U_{cycle} = 0$.

Heat Exchanged (Q): From the first law of thermodynamics, $\Delta U = Q - W$.

For the cycle, $\Delta U_{cycle} = Q_{cycle} - W_{cycle}$.

The work done on the gas is 10 J. This means the work done by the gas is $W_{cycle} = -10\text{ J}$.

$0 = Q_{cycle} - (-10\text{ J}) \Rightarrow Q_{cycle} = -10\text{ J}$.

The negative sign indicates that 10 J of heat is rejected by the gas to the surroundings.

8. First, calculate the acceleration of the body using Newton's Second Law:

$$F = ma \Rightarrow a = \frac{F}{m} = \frac{20\text{ N}}{5\text{ kg}} = 4\text{ m/s}^2.$$

Now, use the first equation of motion to find the final velocity:

$v = u + at$. Since the body starts from rest, $u = 0$.

$v = 0 + (4\text{ m/s}^2)(5\text{ s}) = 20\text{ m/s}.$

9. Let $f = Kr^a d^b s^c$, where K is a dimensionless constant.

Writing the dimensions:

$$[T^{-1}] = [L]^a [ML^{-3}]^b [MT^{-2}]^c [M^0L^0T^{-1}]$$

$$= [M^{b+c}L^{a-3b}T^{-2c}]$$

Comparing the powers of M, L, and T:

- For $M: b + c = 0 \Rightarrow b = -c$

- For T: $-2c=-1\Rightarrow c=\frac{1}{2}$
- For $L: a-3b=0\Rightarrow a=3b=3(-c)=\frac{-3}{2}$

Substituting the values of a, b, and c:

$$f=Kr^{-3/2}d^{-1/2}s^{1/2}=K\sqrt{\frac{s}{r^3d}}.$$

OR

A person stepping out of a moving bus has inertia of motion due to which their body continues to move in the same direction as the bus, even after their feet have touched the ground. If they stop abruptly, their feet come to rest while the rest of their body continues to move forward, causing them to fall. Running for a short distance helps them gradually reduce their body's forward momentum, bringing them safely to rest without falling.

10. The fundamental frequency of an open organ pipe is $f_{open}=\frac{v}{2L}$.

The fundamental frequency of a closed organ pipe is $f_{closed}=\frac{v}{4L}$.

We are given $f_{open}=400\text{ Hz}$.

Since the lengths are the same, we have:

$$f_{closed}=\frac{1}{2}\times\frac{v}{2L}=\frac{1}{2}\times f_{open}=\frac{1}{2}\times 400\text{ Hz}=200\text{ Hz}.$$

11. Horizontal component of velocity:

$$u_x=u\cos\theta=10\cos(30^\circ)=10\times\frac{\sqrt{3}}{2}=5\sqrt{3}\approx 8.66\text{ m/s}.$$

Time of flight,

$$t=\frac{\text{horizontal distance}}{u_x}=\frac{17.3}{8.66}\approx 2\text{s}.$$

Vertical component of velocity:

$$u_y=u\sin\theta=10\sin(30^\circ)=10\times 0.5=5\text{ m/s}.$$

Using the second equation of motion for vertical displacement: $y=u_yt+\frac{1}{2}a_yt^2$.

The displacement y is the height of the tower, and acceleration a_y is $-g$.

$$y=(5)(2)+\frac{1}{2}(-10)(2)^2=10-20=-10\text{ m}.$$

The negative sign indicates the displacement is in the downward direction. So, the height of the tower is 10m.

12. The coefficient of performance (COP) of a refrigerator is given by: $\text{COP}=\frac{Q_2}{W}=\frac{T_2}{T_1-T_2}$.

Given: Cold temperature

$T_2=-3^\circ\text{C}=-3+273=270\text{ K}$.

Room temperature $T_1=27^\circ\text{C}=27+273=300\text{ K}$.

$$\text{COP}=\frac{270}{300-270}=\frac{270}{30}=9.$$

This means 9 joules of heat are removed from the cold reservoir for each joule of work done.

From the first law of thermodynamics,

$Q_1=W+Q_2$.

Heat delivered to the room is Q_1. For each joule of electrical energy consumed ($W=1\text{ J}$), the heat delivered is $Q_1=1\text{ J}+Q_2=1+9=10\text{ J}$.

13.
- Change in momentum (Δp) is equal to impulse (J), where $J=F\Delta t$.

  $\Delta p=(100\text{ N})(10\text{ s})=1000\text{ kg}\cdot\text{m/s}$
- We know that $\Delta p=m\Delta v=m(v-u)$. Since the body starts from rest $(u=0), \Delta p=mv$.

  $1000\text{ kg}\cdot\text{m/s}=(50\text{ kg})v$

  $v=\frac{1000}{50}=20\text{ m/s}$

14. Consider a particle moving in a circular path of radius r. Let it travel a linear distance s in a small time t. The corresponding angular displacement is θ.'

[4]

From the definition of angular displacement,

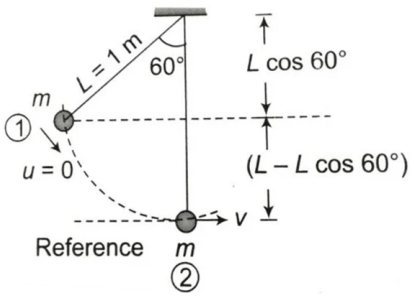
$\theta = \frac{s}{r} \Rightarrow s = r\theta$.

Dividing by time t, we get: $\frac{s}{t} = r\frac{\theta}{t}$.

As $t \to 0, \frac{ds}{dt} = v$ (linear velocity) and $\frac{d\theta}{dt} = \omega$ (angular velocity).

Therefore, $\boldsymbol{v} = r\omega$.

15.



At the mean position, the forces acting on the bob are the tension (T) in the string upwards and the weight (mg) downwards. The net force provides the centripetal force, $T - mg = \frac{mv^2}{L}$.

We need to find the velocity (v) at the mean position. Using the conservation of energy, the loss in potential energy is equal to the gain in kinetic energy.

Loss in PE: $mgh = \frac{1}{2}mv^2 \Rightarrow v^2 = 2gh$.

From the diagram, the height

$h = L - L\cos\theta = L(1 - \cos\theta)$.

$v^2 = 2gL(1 - \cos\theta) = 2(9.8)(1)(1 - \cos 60°)$

$= 2(9.8)(0.5) = 9.8\ m^2/s^2$

Now, substitute v^2 into the tension equation:

$T = mg + \frac{mv^2}{L} = (0.1)(9.8) + \frac{(0.1)(9.8)}{1}$

$= 0.98 + 0.98 = 1.96$ N.

OR

The orbital velocity of a satellite is given by the expression $v = \sqrt{\frac{GM}{r}}$.

When the radius is increased to $r' = 4r$, the new orbital velocity will be:

$v' = \sqrt{\frac{GM}{r'}} = \sqrt{\frac{GM}{4r}} = \frac{1}{2}\sqrt{\frac{GM}{r}} = \frac{1}{2}v$.

Thus, the orbital velocity is halved.

16. Work done by the gas is given by $W = \int_{V_1}^{V_2} PdV$.

For an ideal gas, the equation of state is

$PV = nRT$, so $P = \frac{nRT}{V}$.

$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV$. Since it is an isothermal process, T is constant.

$W = nRT\int_{V_1}^{V_2} \frac{1}{V} dV = nRT[\ln V]_{V_1}^{V_2}$

$= nRT(\ln V_2 - \ln V_1) = nRT\ln\left(\frac{V_2}{V_1}\right)$

17. The total energy of a simple harmonic oscillator is $E = \frac{1}{2}kA^2$, where k is the force constant and A is the amplitude.

The kinetic energy is given by

$KE = \frac{1}{2}m\omega^2(A^2 - x^2)$.

We know that $k = m\omega^2$, so $KE = \frac{1}{2}k(A^2 - x^2)$.

When the displacement is half the amplitude,

$x = \frac{A}{2}$.

$KE = \frac{1}{2}k\left(A^2 - \left(\frac{A}{2}\right)^2\right)$

$= \frac{1}{2}k\left(A^2 - \frac{A^2}{4}\right)$

$= \frac{1}{2}k\left(\frac{3}{4}A^2\right)$

$= \frac{3}{4}\left(\frac{1}{2}kA^2\right) = \frac{3}{4}E$.

The kinetic energy is $\frac{3}{4}E$.

[5]

18. Newton's Second Law of Motion: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts. Mathematically, $F = \frac{dp}{dt}$, where p is the momentum.

Impulse: Impulse is the measure of the effect of a force on a body over a given time interval. It is the product of the force and the time for which it acts. Impulse, $J = F\Delta t$.

Proof:

From Newton's second law, $F = \frac{dp}{dt}$. $Fdt = dp$

Integrating both sides over the time interval from t_1 to t_2 :

$\int_{t_1}^{t_2} Fdt = \int_{p_1}^{p_2} dp$

For a constant force, $F\int_{t_1}^{t_2} dt = p_2 - p_1$

$F(t_2 - t_1) = p_2 - p_1$

$F\Delta t = \Delta p$

Since Impulse $J = F\Delta t$, we have $J = \Delta p$. This proves that impulse is equal to the change in momentum.

OR

1. Law of Orbits: All planets move in elliptical orbits with the Sun at one of the two foci. This law changed the geocentric view of the solar system, providing a foundation for understanding planetary motion.
2. Law of Areas: A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.
   This law is a consequence of the conservation of angular momentum. It means a planet moves faster when it is closer to the Sun and slower when it is farther away.
3. Law of Periods: The square of the orbital period (T) of a planet is directly proportional to the cube of the semi-major axis (r) of its orbit. $T^2 \propto r^3$. This law provides a quantitative relationship between a planet's orbital period and its distance from the Sun. It was a key piece of evidence for Newton's law of universal gravitation.

19.

- Coefficient of Linear Expansion (α): Defined as $\alpha = \frac{\Delta L}{L\Delta T}$
- Coefficient of Superficial Expansion (β): Defined as $\beta = \frac{\Delta A}{A\Delta T}$
- Derivation of $\beta = 2\alpha$: Consider a square plate of side L. Initial area is $A = L^2$. When heated, the new length is $L' = L(1 + \alpha\Delta T)$.
  New area is $A' = (L')^2 = [L(1 + \alpha\Delta T)]^2$
  $= L^2(1 + 2\alpha\Delta T + \alpha^2\Delta T^2)$.
  Neglecting higher order terms,
  $A' \approx L^2(1 + 2\alpha\Delta T) = A(1 + 2\alpha\Delta T)$.
  Comparing this with $A' = A(1 + \beta\Delta T)$, we get $\beta = 2\alpha$.
- Coefficient of Cubical Expansion (γ): Defined as $\gamma = \frac{\Delta V}{V\Delta T}$
- Derivation of $\gamma = 3\alpha$: Consider a cube of side L. Initial volume is $V = L^3$. When heated, the new length is $L' = L(1 + \alpha\Delta T)$.
  New volume is
  $V' = (L')^3 = [L(1 + \alpha\Delta T)]^3 = L^3(1 + 3\alpha\Delta T + \ldots)$
  . Neglecting higher order terms,
  $V' \approx L^3(1 + 3\alpha\Delta T) = V(1 + 3\alpha\Delta T)$.
  Comparing with $V' = V(1 + \gamma\Delta T)$, we get $\gamma = 3\alpha$.
- Final Relationship: Therefore, $\alpha : \beta : \gamma = 1 : 2 : 3$.

[6]

OR

Consider a simple pendulum of length L and a bob of mass m. When the bob is displaced by a small angle θ from the equilibrium position, the restoring force is the component of gravity actinc along the arc: $F = -mg\sin\theta$.

For small angles, $\sin\theta \approx \theta$. So, $F = -mg\theta$.

The arc length is $x = L\theta$, so $\theta = \frac{x}{L}$.

Substituting this into the force equation:

$F = -mg\frac{x}{L}$.

This is in the form of Hooke's Law, $F = -kx$,

where the force constant $k = \frac{mg}{L}$.

The time period of a simple harmonic oscillator is given by $T = 2\pi\sqrt{\frac{m}{k}}$.

Substituting the value of

$k : T = 2\pi\sqrt{\frac{m}{mg/L}} = 2\pi\sqrt{\frac{L}{g}}$.

This is the expression for the time period of a simple pendulum.

20. (a)

- Power: Power is defined as the rate at which work is done or energy is transferred. Mathematically, $P = \frac{W}{t}$. The SI unit of power is the watt (W).
- Average Power: It is the total work done divided by the total time taken to do that work. $P_{avg} = \frac{\Delta W}{\Delta t}$.
- Instantaneous Power: It is the power at a particular instant in time. It is defined as the derivative of work with respect to time, or the product of the force and the velocity at that instant. $P_{inst} = \frac{dW}{dt} = F \cdot v$.

(b)

- Work done: The work done by the pump is equal to the work done against gravity to lift the water.

$W = \text{Force} \times \text{Displacement} = mg \times h$

$W = (100\ \text{kg})(10\ \text{m/s}^2)(20\ \text{m})$

$= 20000\ \text{J}$

- Power of the pump: Power is the rate of doing work.

$P = \frac{W}{t} = \frac{20000\text{J}}{5\text{s}} = 4000\text{W or } 4\text{kW}$

■■■

[7]

SAMPLE QUESTION PAPER

SESSION 2025-2026

PHYSICS

Class 11th

SET-03

Time: 120 Minutes

M.M.: 70

General Instructions:

1. Answers to this Paper must be written on the paper provided separately.
2. You will not be allowed to write during first 15 minutes.
3. This time is to be spent in reading the question paper.
4. The time given at the head of this Paper is the time allowed for writing the answers.
5. Section A is compulsory. Attempt any four questions from Section B.
6. The intended marks for questions or parts of questions are given in brackets [ ].

SECTION-A (14 Marks)

(Attempt all questions from this Section.)

1. In questions (i) to (vii) below, choose the correct alternative (a), (b), (c), or (d) for each of the questions given below:

(i) A physical quantity is measured as 5.00 kg. How many significant figures does this measurement have?

(A) 1
(B) 2
(C) 3
(D) 4

(ii) A body is projected vertically upwards with a velocity u . The time taken to reach the maximum height is:

(A) u / g
(B) $2u / g$
(C) $u^2 / 2g$
(D) $2u^2 / g$

(iii) The work-energy theorem is a direct consequence of which law?

(A) Newton's first law of motion
(B) Newton's second law of motion
(C) Newton's third law of motion
(D) Law of conservation of momentum

(iv) Which of the following is an example of a conservative force?

(A) Frictional force
(B) Air resistance
(C) Viscous force
(D) Gravitational force

(v) In a refrigerator, the working substance extracts heat from the cold reservoir and rejects it to the hot reservoir. This is a process that:

(A) Violates the first law of thermodynamics

(B) Violates the second law of thermodynamics

(C) Obeys the first law of thermodynamics but not the second

(D) Obeys both the first and second laws of thermodynamics

(vi) The angular momentum of a particle about a point is conserved if:

(A) The net force on the particle is zero

(B) The net torque on the particle about that point is zero

(C) The linear momentum of the particle is conserved

(D) The kinetic energy of the particle is conserved

(vii) The dimensional formula for surface tension is:

(A) $M^1L^0T^{-2}$

(B) $M^1L^1T^{-2}$

(C) $M^1L^{-1}T^{-2}$

(D) $M^0L^1T^{-2}$

(B) Answer the following questions briefly:

(i) State the principle of homogeneity of dimensions.

(ii) Define the term 'centripetal acceleration'.

(iii) State the law of conservation of energy.

(iv) What is the physical significance of the area under a velocity-time graph?

(v) State the Zeroth Law of Thermodynamics.

(vi) Give the relationship between the coefficient of linear expansion and the coefficient of cubical expansion.

(vii) What is a geostationary satellite?

SECTION B – 14 MARKS

2. Define the term 'streamline flow' and 'turbulent flow'.

OR

State the principle of superposition of waves.

3. State the parallelogram law of vector addition. Derive an expression for the magnitude and direction of the resultant vector.

4. Derive the expression for the orbital velocity of a satellite.

OR

State and prove the Work-Energy Theorem for a variable force.

5. State the first law of thermodynamics. Explain how it relates heat, work, and internal energy.

6. Define the moment of inertia. State the parallel and perpendicular axes theorems

7. An ideal gas is taken through a cycle $A \rightarrow B \rightarrow C \rightarrow A$. The work done on the gas during the cycle is 10 J.

- What is the change in internal energy during the cycle?
- What is the heat exchanged during the cycle?

8. A body of mass 5 kg is initially at rest. A constant force of 20 N acts on it for 5 seconds. Calculate the final velocity of the body.

SECTION C – 27 MARKS

9. The frequency f of an oscillatory liquid drop depends on the radius r of the drop, density d, and surface tension s. Obtain the formula for frequency f using dimensional analysis.

OR

Explain why a person stepping out of a bus in motion should run in the same direction for a short distance.

10. A closed organ pipe and an open organ pipe are of the same length. If the open organ pipe produces a fundamental frequency of 400 Hz, what is the fundamental frequency of the closed organ pipe?

11. A ball is thrown from the top of a tower with an initial velocity of 10 m / s at an angle of $30°$ above the horizontal. It hits the ground at a horizontal distance of 17.3 m from the base of the tower. Calculate the height of the tower. ($g = 10\ m/s^2$)

12. The temperature inside a refrigerator is $-3°C$ and the room temperature is $27°C$. How many joules of heat will be delivered to the room for each joule of electrical energy consumed ideally?

13. A force of 100 N acts on a body of mass 50 kg for 10 seconds. Calculate the change in momentum of the body and its final velocity, assuming it started from rest.

14. Derive the relation between linear velocity and angular velocity.

15. A pendulum of length 1 m has a bob of mass 100 g. It is displaced to an angle of $60°$ from the vertical. What is the tension in the string when the bob passes through its mean position?

OR

What is the effect on the orbital velocity of a satellite if the radius of its orbit is increased by a factor of 4?

16. A gas expands isothermally from volume V_1 to V_2. Derive the expression for the work done by the gas during this process.

17. A simple harmonic oscillator has a total energy of E. What is its kinetic energy when the displacement is half the amplitude?

SECTION D – 15 MARKS

18. State Newton's second law of motion and define impulse. Prove that impulse is equal to the change in momentum.

OR

State and explain Kepler's laws of planetary motion.

19. Derive the relationship between the coefficient of linear expansion, superficial expansion, and cubical expansion.

OR

Derive the expression for the time period of a simple pendulum.

20. (a) Define power and distinguish between average power and instantaneous power.

(b) A pump is used to lift water from a well of depth 20 m. The pump lifts 100 kg of water in 5 seconds. Calculate the power of the pump. (Assume $g = 10\ m/s^2$)

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